

✓ Armstrong number or not

```
n = int(input("Enter number: "))
digits = len(str(n))
total = sum(int(d)**digits for d in str(n))

if total == n:
    print("Armstrong")
else:
    print("Not Armstrong")
```

Enter number: 13
Not Armstrong

✓ perfect number or not

```
n = int(input("Enter a number: "))
s = sum(i for i in range(1, n) if n % i == 0)

if s == n:
    print("Perfect number")
else:
    print("Not a perfect number")
```

Enter a number: 7
Not a perfect number

✓ Prime number or not

```
n = int(input("Enter a number: "))

if n < 2:
    print("Not a prime number")
else:
    for i in range(2, int(n**0.5) + 1):
        if n % i == 0:
            print("Not a prime number")
            break
    else:
        print("Prime number")
```

Enter a number: 13
Prime number

✓ Check whether 0 given number is positive,negative

```
n = int(input("Enter a number: "))

if n > 0:
    print("Positive number")
elif n < 0:
    print("Negative number")
else:
    print("Zero")
```

Enter a number: 13
Positive number

✓ Check the eligibility for the vote

```
age = int(input("Enter your age: "))

if age >= 18:
    print("You are eligible to vote ")
else:
    print("You are not eligible to vote ")
```

```
Enter your age: 20
You are eligible to vote
```

✓ Check whether leap year or not

```
year = int(input("Enter a year: "))

if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
    print("Leap year ")
else:
    print("Not a leap year ")
```

```
Enter a year: 2005
Not a leap year
```

✓ Simple calculator

```
a = float(input("Enter first number: "))
op = input("Enter operator (+, -, *, /): ")
b = float(input("Enter second number: "))

if op == "+":
    print("Result:", a + b)
elif op == "-":
    print("Result:", a - b)
elif op == "*":
    print("Result:", a * b)
elif op == "/":
    if b == 0:
        print("Division by zero not allowed ")
    else:
        print("Result:", a / b)
else:
    print("Invalid operator ")
```

```
Enter first number: 13
Enter operator (+, -, *, /): *
Enter second number: 15
Result: 195.0
```

✓ Using while loop count no.of even digits in number

```
n = int(input("Enter a number: "))
temp = n
count = 0

while temp > 0:
    digit = temp % 10

    if digit % 2 == 0:
        count += 1
    temp = temp // 10

print("Number of even digits:", count)
```

```
Enter a number: 132
Number of even digits: 1
```

✓ hello reverse program

```
s="hello"
rev=" "
for i in range(len(s)-1,-1,-1):
    rev=rev+s[i]
print(rev)
```

olleh

✓ Anagram program

```
s1 = "listen"
s2 = "silent"

if sorted(s1) == sorted(s2):
    print("Anagram")
else:
    print("Not Anagram")
```

Anagram

✓ Check if a string contains only numbers

```
s = input("Enter a string: ")

flag = 1

for i in s:
    if i < '0' or i > '9':
        flag = 0

if flag == 1:
    print("Only Numbers")
else:
    print("Not Only Numbers")
```

Enter a string: 13
Only Numbers

✓ Find the longest word in the sentence

```
s = input("Enter a sentence: ")

words = s.split()

longest = words[0]

for word in words:
    if len(word) > len(longest):
        longest = word

print("Longest word:", longest)
```

Enter a sentence: I Become a Data Analyst
Longest word: Analyst

✓ Reverse the words in the sentence

```
s = input("Enter a sentence: ")

words = s.split()

for i in range(len(words)-1, -1, -1):
    print(words[i], end=" ")
```

```
Enter a sentence: I Become a Data Analyst
Analyst Data a Become I
```

✓ Counting vowels and consonants

```
s = "fghy57687ijh hy6"

vo = 0
con = 0
dig = 0
space = 0

for ch in s:
    if ch in "aeiouAEIOU":
        vo += 1
    elif ch.isdigit():
        dig += 1
    elif ch == " ":
        space += 1
    else:
        con += 1

print("Vowels:", vo)
print("Consonants:", con)
print("Digits:", dig)
print("Spaces:", space)
```

```
Vowels: 1
Consonants: 8
Digits: 6
Spaces: 1
```

✓ Counting repeated letters

```
word = "Data Analyst"
freq = {}

for ch in word:
    if ch != " ":
        if ch in freq:
            freq[ch] += 1
        else:
            freq[ch] = 1

for key in freq:
    if freq[key] > 1:
        print(key, "=", freq[key])
```

```
a = 3
t = 2
```

```
a = [9, 5, 3, 4, 5, 3, 9, 4]
b = []

for i in a:
    if i not in b:
        b.append(i)

print(sorted(b))
```

```
[3, 4, 5, 9]
```

```
s1 = input("Enter even number of String only example is abcd: ")
s2 = input("Enter even number of String only example is cdab: ")
```

```
if s2 == s1[2:] + s1[:2]:  
    print("True")  
else:  
    print("False")  
  
print(s1[2:]+s1[:2])
```

```
Enter even number of String only example is abcd: bcd  
Enter even number of String only example is cdab: abcd  
False  
dfbc
```

✓ Palindrome

```
s = "A place, paris, germany"  
cleaned = s.replace(", ", "").replace(" ", "").lower()  
print(cleaned)  
s1 = cleaned[::-1]  
if cleaned == s1:  
    print("Palindrome")  
else:  
    print("Not a palindrome")
```

```
aplaceparisgermany  
Not a palindrome
```

Double-click (or enter) to edit

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